Extended reading — Week 01: MDPs and Bandits

October 20, 2026

Read with a question in mind

Start with the week's teaching guide and its small worked example. Then use these readings to connect the algorithm to its assumptions and evidence. The explanations and investigations below are course commentary, not quotations. Read the source before reusing a theorem or a reported result. For each paper, record the objective, observation model, claimed result, conditions and one condition your local experiment does not reproduce.

1. Robbins (1952): Some aspects of the sequential design of experiments

Primary source or book record: https://www.ams.org/journals/bull/1952-58-05/S0002-9904-1952-09620-8/.

What to learn. An early mathematical treatment of sequential allocation with unknown rewards. Read it to identify the decision, the available observations and the comparison with an informed decision maker.

What the result requires. Historical terminology need not match a modern regret definition. Do not infer a logarithmic finite-time guarantee from an asymptotic statement or treat play-the-winner as universally consistent.

Notebook investigation. Write a two-arm model with means 0.4 and 0.6. If the inferior arm is pulled 30 times, expected pseudo-regret is 6; realised rewards need not differ by exactly 6.

2. Auer, Cesa-Bianchi and Fischer (2002): Finite-time Analysis of the Multiarmed Bandit Problem

Primary source or book record: https://homes.di.unimi.it/~cesabian/Pubblicazioni/ml-02.pdf.

What to learn. UCB1 combines empirical means with a bonus depending on sample count. For bounded stationary rewards, the expected pulls of a suboptimal arm scale with $\log T/\Delta_a^2$. Multiplying by the gap gives regret proportional to $\log T/\Delta_a$.

What the result requires. Initialise every arm. In the proof, after enough pulls a suboptimal choice requires a concentration failure; before that threshold the count itself pays the regret bill. A finite-time upper bound is not a prediction of a particular learning curve.

Notebook investigation. Use `exercise_3_ucb1_bonus`. Hold time fixed and quadruple the count: the bonus halves. Compare a multi-seed regret curve with the bound without fitting a claimed exact slope.

3. Agrawal and Goyal (2012): Analysis of Thompson Sampling for the Multi-armed Bandit Problem

Primary source or book record: https://proceedings.mlr.press/v23/agrawal12.html.

What to learn. Analyses Thompson sampling for Bernoulli rewards. The two-arm result is $O(\log T/\Delta + 1/\Delta^3)$; its original general-$K$ result is not the simple UCB1-shaped sum often quoted for later analyses.

What the result requires. Distinguish frequentist regret with fixed means from Bayesian regret averaged over a prior. This theory paper is not evidence for an 18-month Yahoo deployment or a universal practical ranking of algorithms.

Notebook investigation. In `exercise_4_thompson_posterior`, update $\text{Beta}(1,1)$ with three successes and one failure: $\text{Beta}(4,2)$ has mean $2/3$. A posterior draw is not an action propensity.

4. Li, Chu, Langford and Schapire (2010): A Contextual-Bandit Approach to Personalized News Article Recommendation

Primary source or book record: https://arxiv.org/abs/1003.0146.

What to learn. Introduces disjoint and hybrid LinUCB models and evaluates contextual recommendation using randomized news logs. The paper reports a 12.5 percent relative click improvement in its evaluation; it does not establish a doubled click rate sustained for 18 months.

What the result requires. Uniform random logging supports replay under the stated evaluation assumptions. Sharing coefficients can help when information transfers across arms, but can hurt under misspecification. Online updating is not immunity to drift.

Notebook investigation. Compare `exercise_7_linucb_regret` with the noncontextual bake-off. Explain which coefficients are shared and which observations update them. Keep simulated reward gains separate from the paper result.

5. Russo and Van Roy (2014): Learning to Optimize via Information-Directed Sampling

Primary source or book record: https://arxiv.org/abs/1403.5556.

What to learn. Balances squared expected regret against information gained about the optimum. With an appropriate information-ratio bound, Bayesian cumulative regret is controlled by prior entropy and horizon.

What the result requires. IDS minimises a regret-to-information ratio; it does not merely maximise information gain. Bayesian bounds and gap-dependent frequentist UCB bounds answer different questions. An information quantity may be expensive to compute.

Notebook investigation. Construct an action that reveals much about an irrelevant parameter but little about the best action. Explain why exploration should value decision-relevant information.

Synthesis: evidence you should be able to show

1. Reproduce one numerical calculation before running a long experiment.
2. Name the precise objective, sampling distribution and terminal convention.
3. Compare the local implementation with one source: list a shared mechanism and one simplification that prevents calling it a full reproduction.
4. Separate an algebraic identity, an assumption-dependent theorem and an empirical result. A plot alone does not prove a theorem.
5. Save the notebook with seed, parameters, axes, denominators and a short explanation of an unexpected result. Keep evaluation data separate from tuning.

Return to the teaching guide's notebook investigation for the local function names, expected checks and interpretation questions. Source code and notebooks are available together in the course repository; the web reader is the entry point and the downloaded notebook is the execution workspace.